

BIO310 Introduction to Bioinformatics
Lab 6
Spring 2024

April 16, 2026

Instructions:

- Do not forget to sign the attendance sheet, or your work will not be graded.
- This is a lab work, please submit until the end of the lab.
- Submit a PDF document for the answers to the write-up questions. (If present) The plots should be appropriately labeled, and the figures should have captions and be appropriately cited within the main text. Name your submission as **BI0310-LabX-YourName.pdf** where you substitute the lab number instead of X and your first and last names in place of 'YourName'. Submit online through SUCourse as a single file.
- Upload the code (as **.ipynb**) online on SuCourse. Name your submission **BI0310-LabXcode-YourName.ipynb** where you substitute your first and last names into the file name in place of 'YourName' and X with the current lab number.
- Please follow the submission instructions, not adhering the submission standards will lead to point deduction.

Task 1: Exercise: de Bruijn Graphs

1. Given the following 3-mer composition, construct the corresponding de Bruijn graph (draw and show all nodes and edges).
ABO, ALL, BAB, BAL, BOO, BOO, LLO, LOO, NBA, NBA, ONB, ONB, OON, OON, OON
 - (a) Find all possible sequences that can generate this 3-mer composition.
2. Now consider the sequences you found above.
 - (a) Construct the de Bruijn graph for each sequence using $k = 3$. What do you observe? Can the sequences be distinguished at this value of k ?
 - (b) Construct the de Bruijn graph for each sequence using $k = 5$.
 - (c) Compare the $k = 3$ and $k = 5$ graphs. Explain why increasing k resolves (or does not resolve) the ambiguity between the sequences.

Task 2: Assemble the genome of *Veysonella kayaomans*

A group of scientists created a new strain of bacteria to combat climate change, but it mutated into a highly contagious and deadly strain during testing. The bacteria spread rapidly, attacking crops and critical infrastructure, leading to widespread chaos and destruction. Despite efforts to find a cure, the world's population is decimated, and civilization as we know it comes to an end. You are one of the last bioinformaticians, and a biochemist in your team found a way to sequence the genome of the latest strain. This is your final chance to assemble its genome to develop a new cure.

1. You have a Colab Notebook where you store your code for genome assembly. Check [Colab Notebook](#) and also download the `reads-lab6.txt` from SuCourse under Lab-6 -which contains sequencing reads-, and import it to the Colab Notebook.
2. When you check your code, you see that the bacteria deleted some of the essential parts.
3. You need to rebuild `rc` function to reverse complement a DNA string. Because, in order to assemble a genome, you need to consider both strands.
4. You need to rebuild `getkmers` and `DeBruijn` functions so that you can find a path from given reads to construct the genome in `genomeAssembly` function, which is already present.
5. The good thing is bacteria did not eat up your function comments.
6. Report the length of the assembled genome and the integer k that is used.
7. Hurry up, the world needs you.